

GRAPH MACHINE LEARNING

Topologic Memorial

A Reading of Wartime Loss in Kyiv

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Buildings as Graphs

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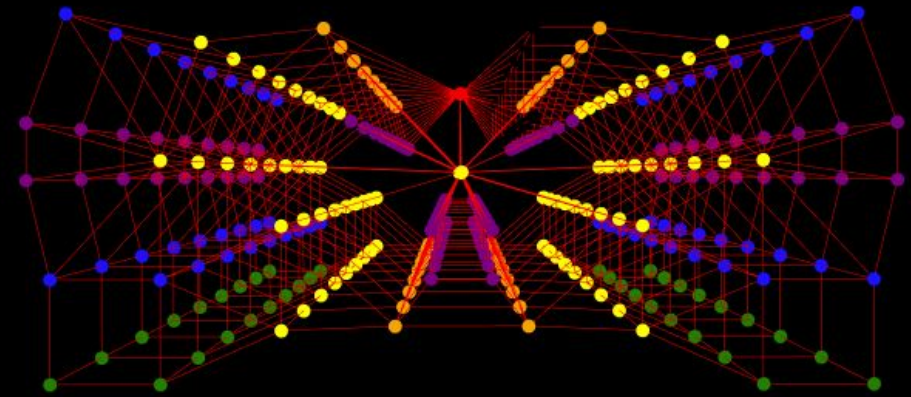
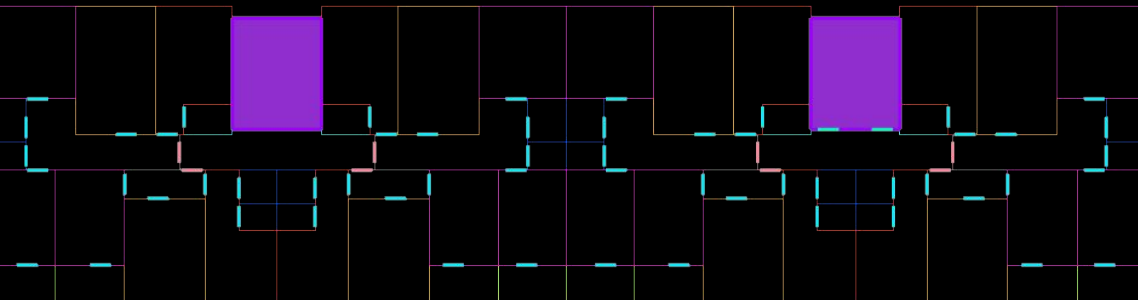
The Building

87-02 Soviet panel block-section. Nine storeys, two apartments per floor, mirrored around a single staircase core. A standard type mass-produced across the USSR from the 1960s onward.

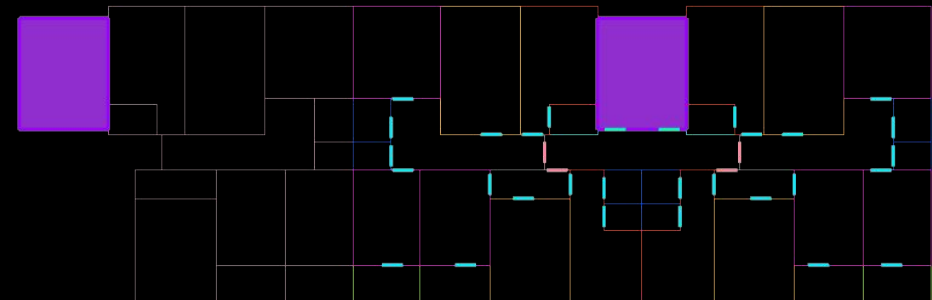
A rocket strike destroyed the right half of one such section in Kyiv. The staircase survived. The apartments on one side did not. The stair now opens onto a concrete void.

The surviving stair leads, in graph terms, to nothing.

The project reads the lost apartment as a spatial graph and uses that graph to generate a memorial form. Each room that no longer exists is given back a floor slab, positioned by the room's topological distance from the entrance.

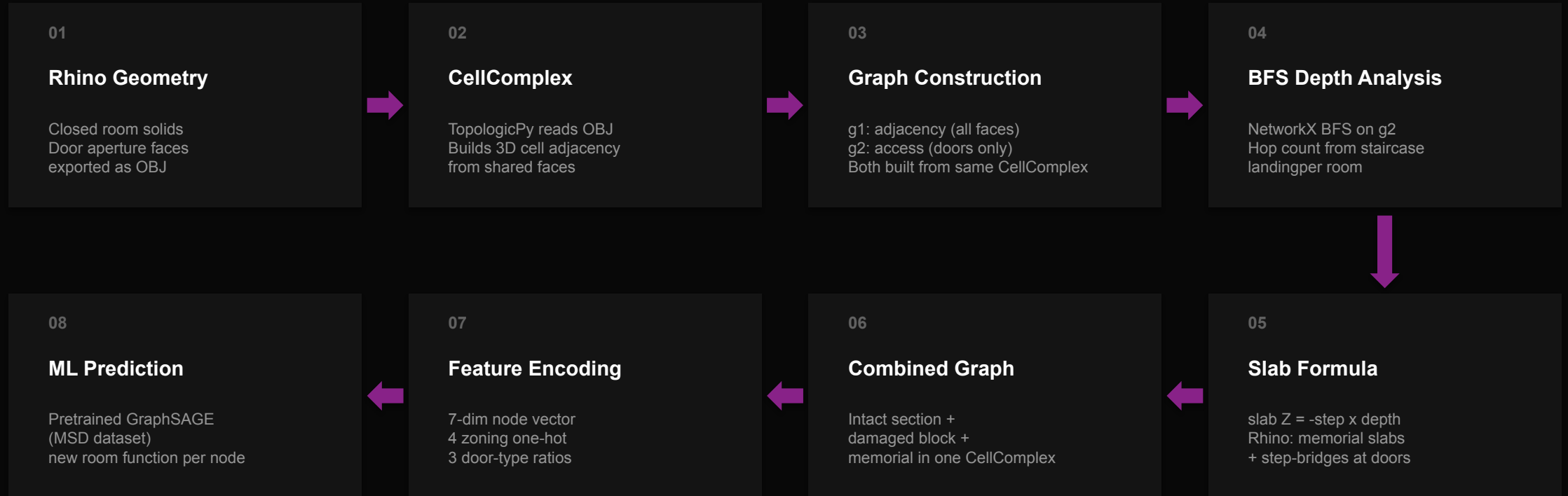


existing typical block adjacency graph



existing plan

Analysis Pipeline



Output: 90 corridor + 63 kitchen + 9 livingroom -- predicted new functions for 162 surviving rooms

Memorial in Topology

Depth logic

The access graph of the damaged section encodes which rooms were adjacent via which door. BFS hop count from the staircase landing assigns a depth to each room on that graph. That depth maps directly to a vertical offset: slab $Z = -\text{step} \times \text{depth}$.

A room at depth 1 -- the apartment hall, directly behind the entrance door -- returns as a slab flush with the landing. A room at depth 3 -- a bedroom, a balcony -- sits three steps below. The further from the entrance, the lower the slab.

In the damaged block, depth ranges from 0 to 12. The promenade descends through 13 levels, each level corresponding to one door crossing in the original apartment.

$$\text{slab } Z = -\text{step} \times \text{depth}$$

Bridge logic

Two connected rooms at different depths sit at different slab heights. The door that connected them in the original apartment marks the position of a step-bridge. The height of that step equals the height difference between the two slabs -- which equals one step unit, because adjacent rooms differ by exactly one BFS hop.

Where two rooms shared a face that was not a door, the gap between slab edges is left open. The opening is the missing wall.

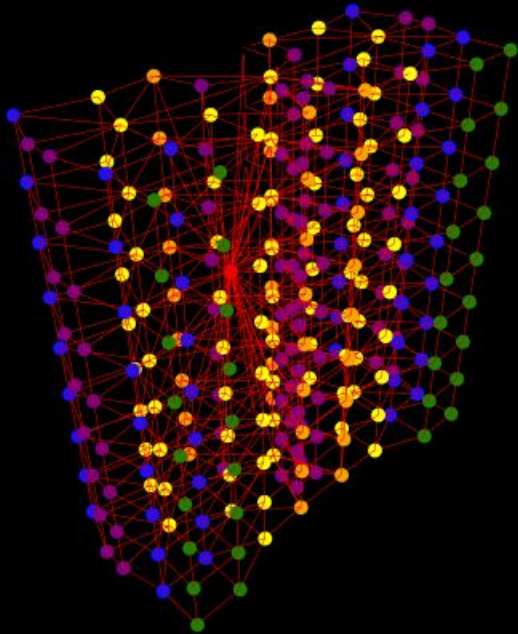
Slab perimeters are shrunk inward by the missing wall thickness. The gap between adjacent slab edges equals that wall dimension. The original room dimensions are still readable in the plan.

The step-bridges are encoded as stairs nodes in the combined graph. The memorial slabs are encoded as balcony nodes. Both are read by the ML model as part of the spatial context when it predicts what the surviving rooms become.

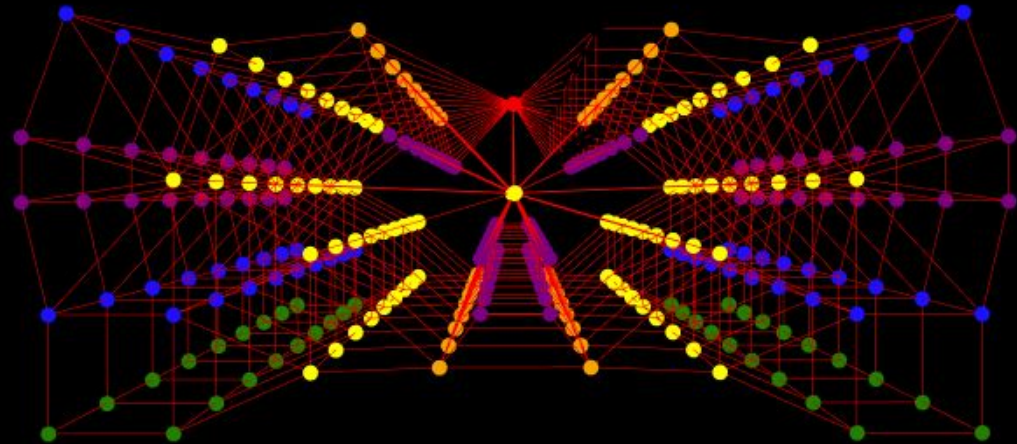
Adjacency Graph g1

`_direct = True` -- every shared face creates an edge, including floors and ceilings between storeys.

The staircase box touches every floor slab in the building; in g1 it connects to every room above and below it, making it an enormous hub. A bedroom on floor 3 has a direct edge to the bedroom on floor 4 through the shared slab. g1 is used for geometry inspection only -- its connectivity is structurally misleading for analysis.



3D isolated -- dense through-slab edges

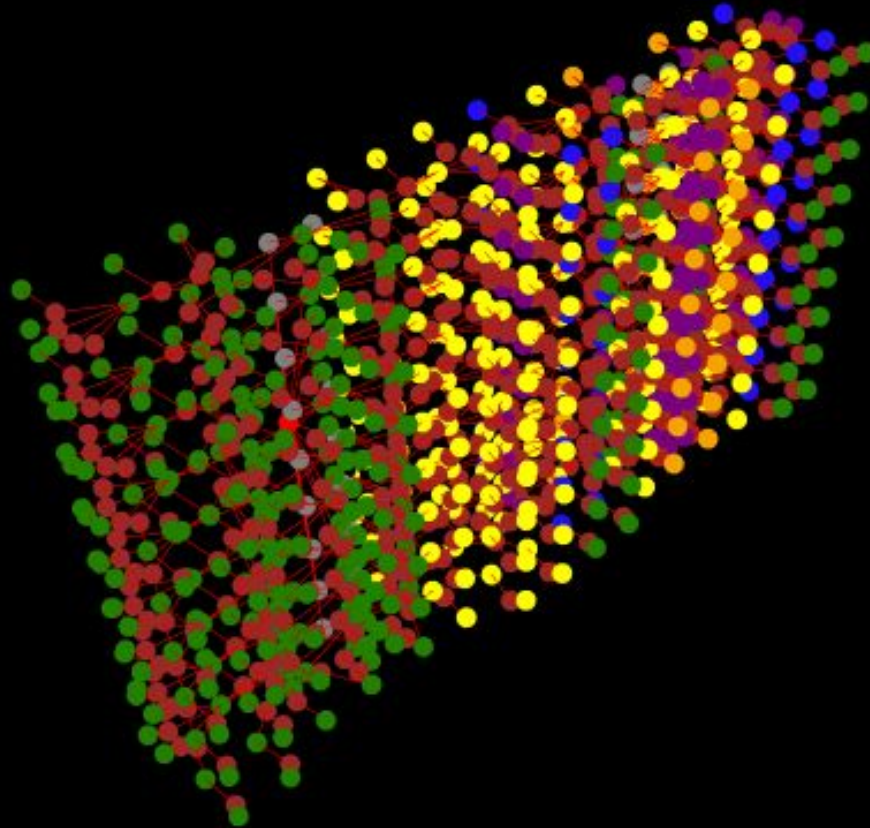


Plan -- repeating double fan, 9 floors collapsed

Access Graph g2

`_direct = False, _viaSharedApertures = True` -- only door aperture faces create edges.

The door OBJ files tag each aperture face by type: entrance door, room door, or passage threshold. Only these faces produce edges. Through-slab connections disappear entirely. Bedrooms and bathrooms become leaf nodes with a single incident edge. The staircase connects only to its landings. g2 represents the movement graph -- the paths a person can actually walk. This is the graph used for BFS depth analysis, slab height computation, feature encoding, and ML training.



Depth from Entrance

BFS runs on g2, starting from the staircase landing node. The hop count to each room is its depth: the number of door apertures a person must cross to reach that room from the staircase. No geometry involved -- only graph structure.

0

Staircase landing

Graph origin. BFS starts here.

1

Apartment hall / corridor

One door crossing: the entrance door.

2

Kitchen, bathroom, storeroom

Two crossings: entrance + one internal door.

3

Bedroom, livingroom, balcony

Three crossings: entrance + two internal doors.

Formula

**slab Z =
-step x depth**

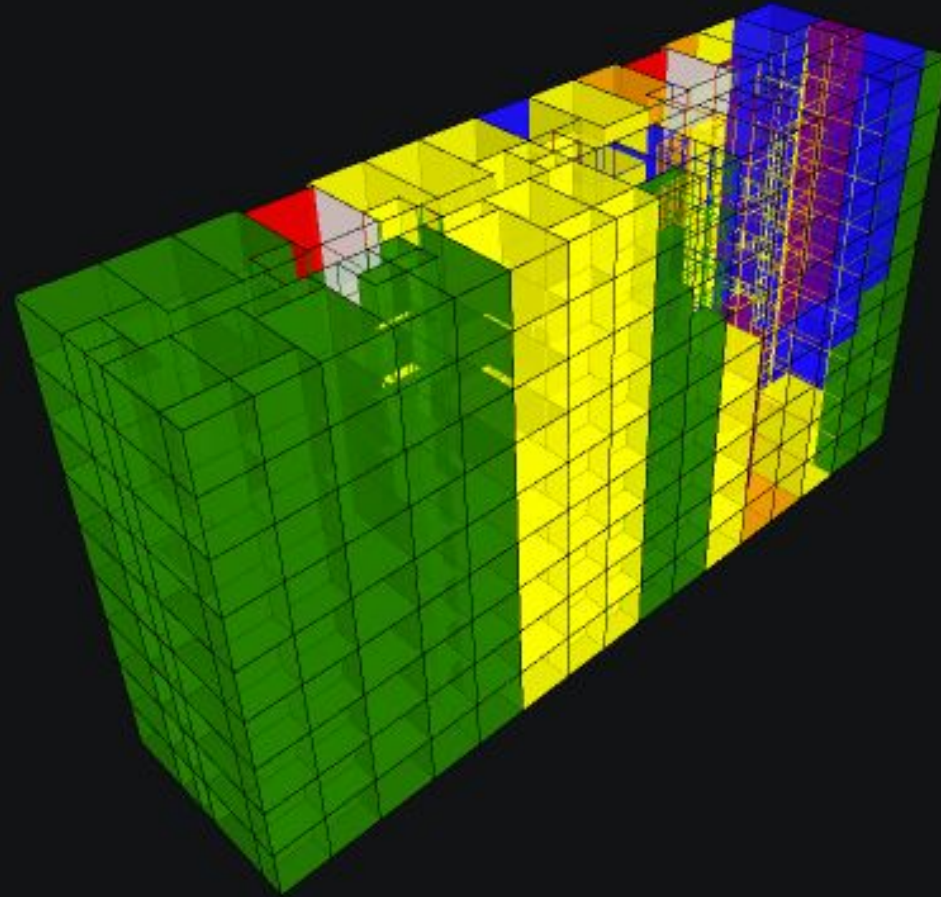
depth 0 --> ground level
depth 12 --> lowest slab
13 distinct Z levels

Intact section

BFS depth ranges from 0 to 3 across the intact nine-storey section. The damaged block, once merged, extends this to depth 12 because the memorial slabs fan outward from the surviving corridors, adding more graph distance from the staircase.

Intact section + Damaged block + Memorial intervention

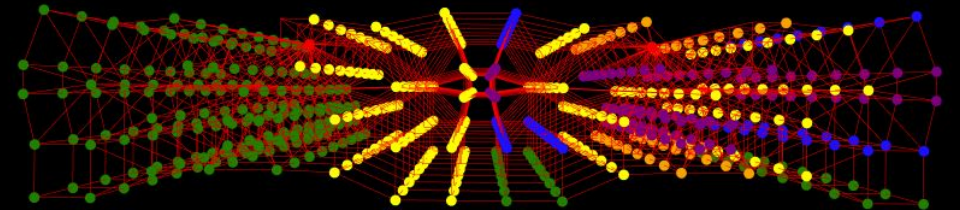
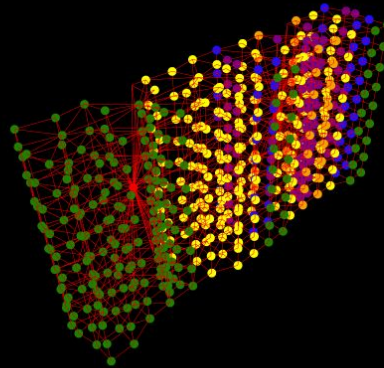
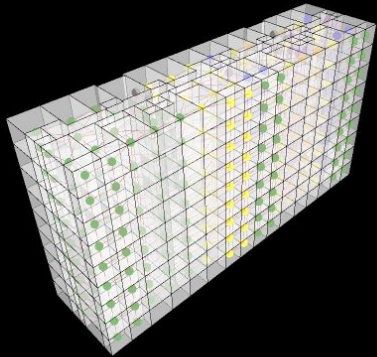
596 cells. The staircase spine at center connects both halves. Right: intact section, real room types. Left: surviving damaged rooms (corridor, yellow) and memorial slabs (balcony, green).



Adjacency Graph g1 -- Both Sections

Through-face adjacency across the full 596-node graph.

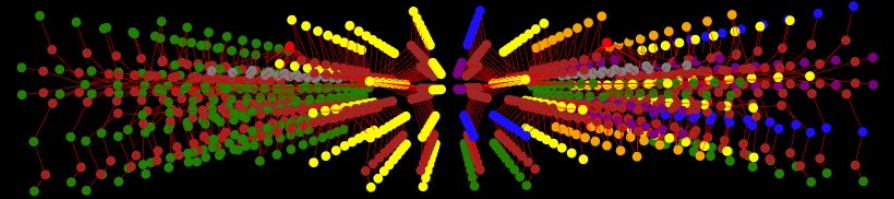
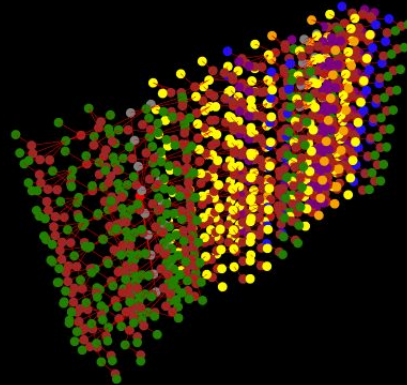
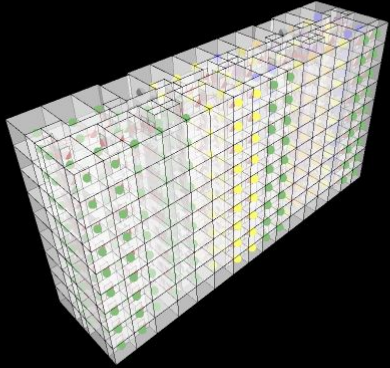
The asymmetry between the two halves is directly visible. The intact section (right) produces the expected double-fan pattern: two apartment clusters per floor, repeating nine times, converging at the staircase. The damaged section (left) produces a different structure: the green balcony slab nodes form a wider, sparser cloud at the periphery -- a typology that did not exist in the original building.



Access Graph g2 -- Both Sections

596 nodes, 594 edges. The staircase landing is the only connection between the two sections.

The intact section (right) reads as a standard apartment tree: staircase branching to landings, landings to entrance doors, halls to rooms, rooms ending as leaves. The damaged section (left) carries the same branching logic but through different node types: surviving corridor nodes fan to balcony slab leaves, with stair-bridges at the original door positions. The lost apartment's topology is structurally intact -- the spatial relationships remain, even though the rooms are gone.



GraphSAGE -- Feature Design

The classifier (pretrained on the Modified Swiss Dwellings, ~1000 apartment plans) takes a 7-dimensional feature vector per node. No coordinates, no area, no geometry.

Room labels and zoning assumptions

The nine room types follow the Modified Swiss Dwellings / CVAAD challenge class order:

room_type	label
bedroom	0
livingroom	1
kitchen	2
dining	3
corridor	4
stairs	5
storeroom	6
bathroom	7
balcony	8

Node feature vector x in $\mathbb{R}^7$

zoning_type_0

1 if private/static (bedroom)

zoning_type_1

1 if living/dynamic (kitchen, corridor, dining)

zoning_type_2

1 if service/functional (bathroom, storeroom, stairs)

zoning_type_3

1 if outdoor (balcony)

connectivity_0

fraction of edges that are passage thresholds

connectivity_1

fraction of edges that are room doors

connectivity_2

fraction of edges that are entrance doors

The four zoning features are one-hot. The three connectivity features are ratios computed from the edge set of each node. A corridor node in the damaged section is assigned zoning_type_1 = 1. Its connectivity features depend on which door types connect it to its neighbors.

Assigned Labels

Before the model runs, every node in the combined graph must have a label. The intact section nodes carry their real room types. The memorial nodes are assigned labels that encode the design intent.

162 surviving damaged rooms	corridor	zoning [0,1,0,0]	Neutral placeholder. zoning_type_1 = 1. The model will predict the new function for each of these nodes.
216 memorial slabs	balcony	zoning [0,0,0,1]	Outdoor / semi-outdoor. Signals open space to neighboring nodes through the aggregation step.
2 step-bridges	stairs	zoning [0,0,1,0]	Service / functional. Encodes the passage relationship between slabs at different heights.

bedroom

livingroom

kitchen

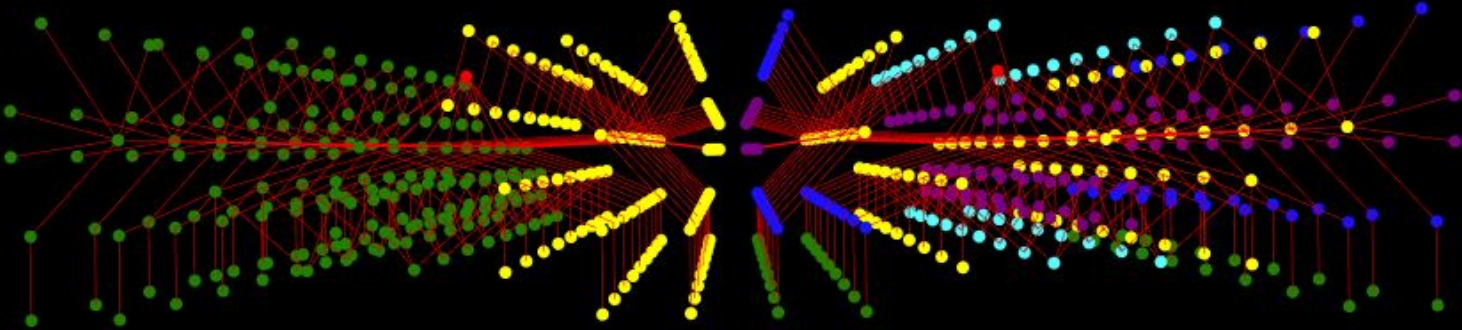
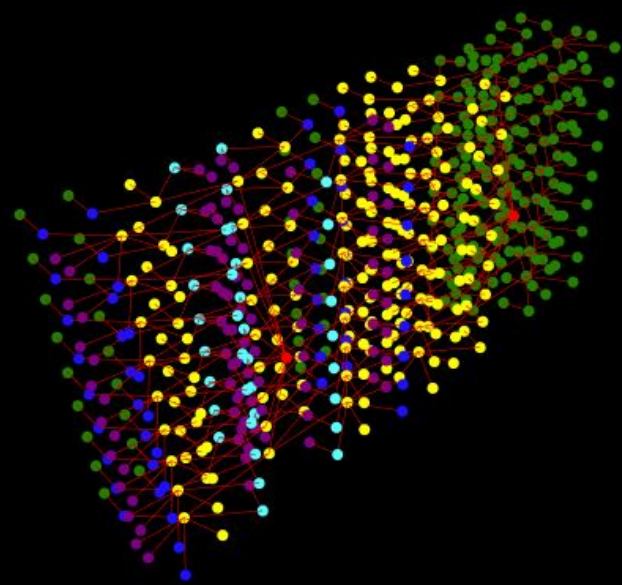
corridor

stairs

storeroom

bathroom

balcony



3D view -- yellow = corridor placeholder (162), green = balcony slabs (216)

Plan view -- the uniform assignment in the damaged section is explicit

How GraphSAGE Classifies Nodes

1. Neighborhood sampling

For each node, GraphSAGE samples a fixed number of neighbors at each layer. In this graph: 1-hop neighbors are rooms directly connected by a door aperture.

2. Feature aggregation

The sampled neighbors' feature vectors are aggregated (here: mean). The aggregated vector is concatenated with the node's own vector and passed through a learned linear transformation. This produces a new embedding for the node.

3. Two-layer propagation

The process runs twice (two GraphSAGE layers), so each node effectively sees its 2-hop neighborhood. In a typical apartment, 2 hops from a corridor reaches bedrooms and balconies -- the full local context of a room.

4. Classification

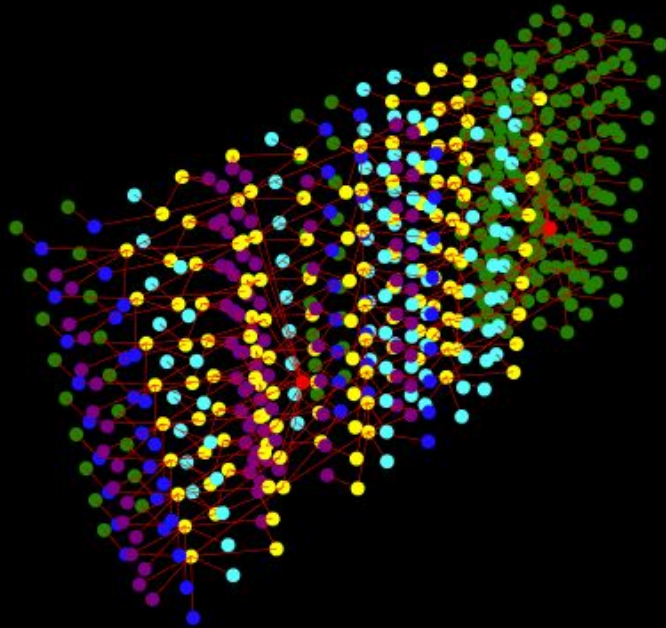
The final embedding is passed through a softmax layer that outputs a probability distribution over the 9 room types. The predicted label is the highest-probability class.

The model was pretrained on MSD (Milan Spatial Dataset) and has not seen this building. It transfers learned connectivity patterns from real apartments to the memorial graph.

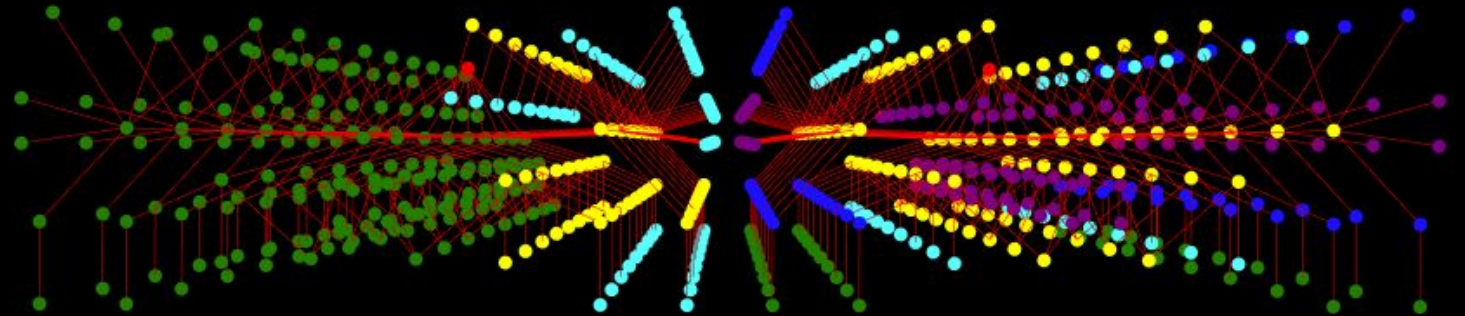
Predicted Functions

After two-layer GraphSAGE aggregation, the corridor nodes in the damaged section carry updated embeddings that reflect their local neighborhood -- which now includes balcony slab nodes and stair-bridge nodes introduced by the design.

Corridor nodes at depth 2, with 2-3 door edges and at least one balcony neighbor, produce embeddings that the classifier maps to kitchen. The model's learned kitchen profile -- rooms with mixed door/outdoor connectivity at depth 2 -- matches the connectivity structure of these nodes in the combined graph. 72 of 162 corridor placeholders are reclassified.



3D view -- cyan = kitchen nodes in the damaged section



Plan view -- kitchen predictions concentrate near the staircase hub

Predictions for 162 Surviving Rooms

90

corridor

Rooms at depth 3 or greater. Low degree -- typically 1-2 door edges. Neighbors are other corridor or balcony nodes, without the mixed connectivity that would shift the embedding toward kitchen. The classifier keeps them as circulation.

63

kitchen

Rooms at depth 2, with 2-3 door edges and at least one direct balcony neighbor. The connectivity_1 (door ratio) and zoning_type_1 features, combined with the aggregated outdoor signal from the balcony nodes, produce an embedding the model assigns to kitchen.

9

livingroom

Rooms with a single door edge leading to a larger corridor or balcony node. Lower connectivity_1 ratio, higher degree on the balcony side. The model interprets this single-opening-to-open-space pattern as livingroom.

What the result means

The memorial slabs were labelled balcony because they are open, unenclosed surfaces. That label carried an outdoor zoning signal into the feature vectors of every adjacent corridor node via the aggregation step. The classifier read that signal and assigned kitchen -- the room type most often adjacent to a balcony in the MSD training data. The design choice (slab = balcony) directly produced the ML result (corridor --> kitchen).

Graph analysis as design driver

The graph generates the form, and the form — re-embedded as nodes — reshapes what the model predicts for the surviving rooms.

01

Build g2

access graph from door apertures

02

BFS depth

hop count from the landing

03

Depth → height

slab Z = - step × depth

04

Encode memorial

slabs as balcony, bridges as stairs

05

GraphSAGE

balcony signal aggregated into corridors

06

63 → kitchen

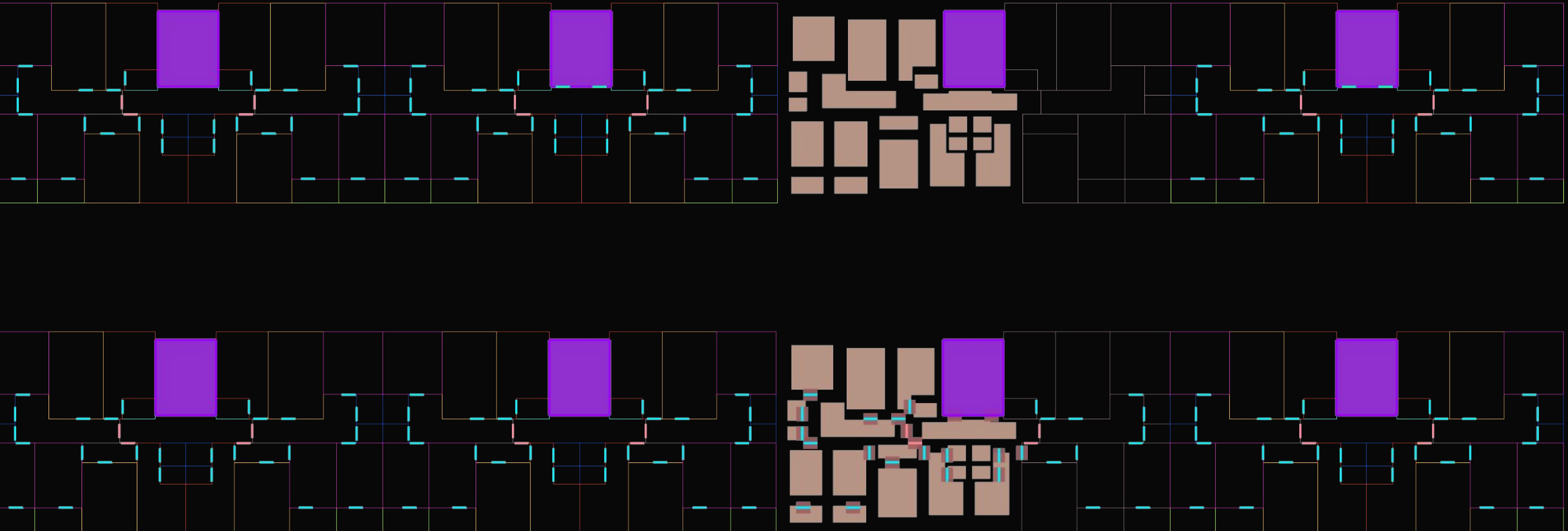
the memorial reads as habitable

A design analysed by a model trained on how people actually live.

Proposed Plan

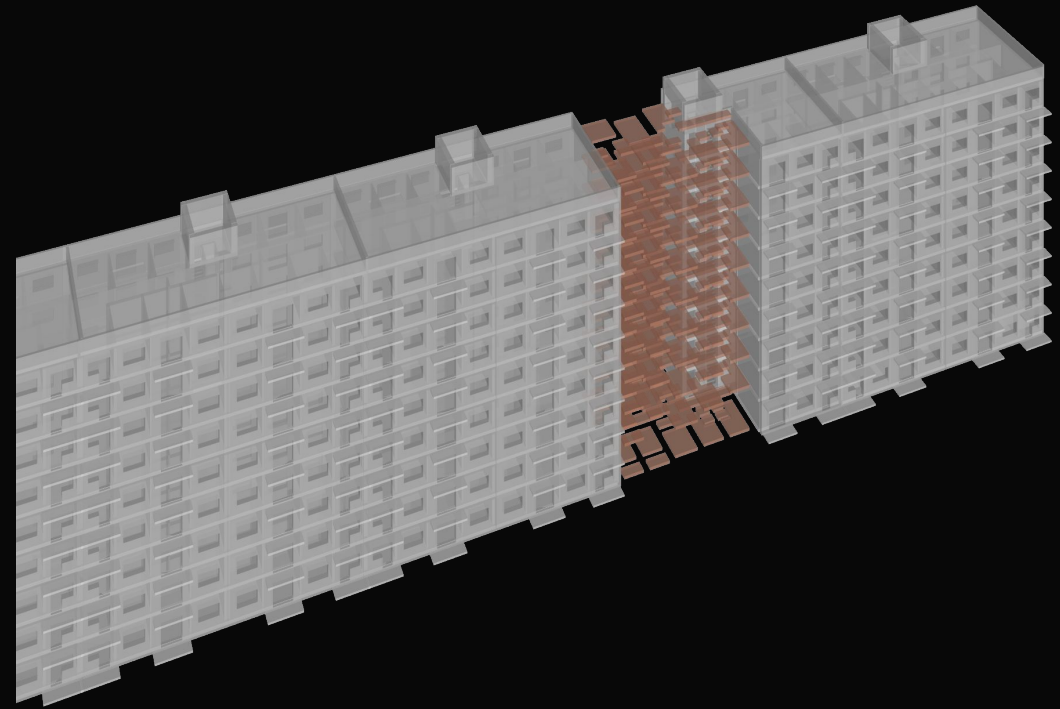
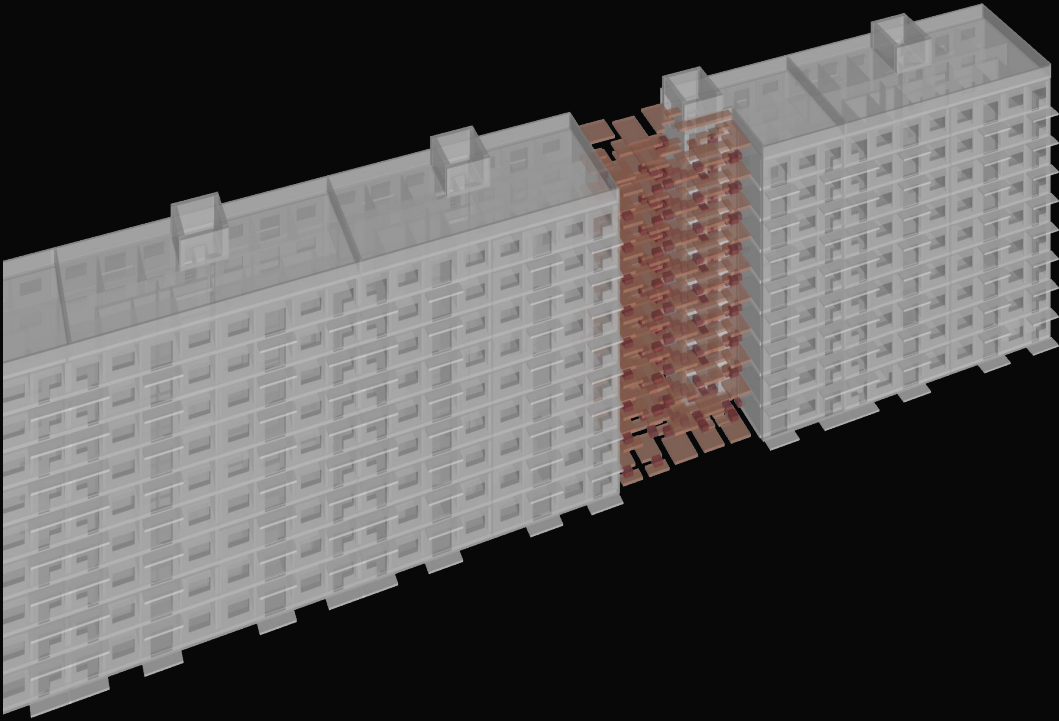
Each destroyed room returns as a slab at the room's original footprint, shrunk inward by the wall thickness. Z-height = $-\text{step} \times \text{BFS depth from entrance}$. The promenade descends from the landing as graph depth increases.

Two connected slabs at different depths sit at different Z levels. The door position between them in the original plan becomes the step. The step height equals one step unit, because adjacent rooms differ by exactly one BFS hop. Slab edges at non-door positions are left open -- the gap is the missing wall.



Memorial Intervention in Context

The depth-mapped slabs step down into the gap. The intact section is unchanged. The staircase at center now leads into a descending spatial sequence that traces the access graph of the lost apartment.



The promenade is the access graph of the lost apartment, made walkable.

Slabs at depth-mapped Z heights. Gaps between slab edges at non-door
positions = the vanished walls. Bridges at door positions connecting slabs at
adjacent depths.

